Vs30: Proxy for Seismic Amplification?

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INTRODUCTION

The seismic codes of many countries base their site classifications on the Vs30 parameter, i.e., the average shear-wave velocity in the first 30 m of subsoil. And yet, despite the widespread use of this parameter, there is not universal agreement that this is a valid proxy for seismic amplification. There is no doubt that since the shear-wave subsoil velocity (Vs) is tied to soil rigidity, the profile of the former must have a role in site effects (Shearer and Orcutt 1987; Cranswick *et al.* 1990). However, seismic amplification appears too complex to be related to the Vs profile in the first 30 m alone. Harmsen (1997) showed largely scattered Vs30 site amplification plots and concluded that source directivity and topography also may play a role in site amplification. Hartzell *et al.* (2001) found a correlation between Vs30 and site amplification, but their conclusion is based on only four Vs30 values (measured only down to 16–17 m and extrapolated to 30 m), all falling in areas of little or no amplification. They did not perform a statistical correlation based on several data equally distributed among all site classes. Boore (2004) concluded that the National Earthquake Hazard Reduction Program (NEHRP) soil classes are a useful way of grouping sites according to site amplification. However, Boore's work involved only class C and D sites, with one exception. Wald and Mori (2000) and Mucciarelli and Gallipoli (2006) raised doubts about the validity of Vs30 as a proxy to soil amplification in tectonically active complex regions or in particular geological settings, such as in the presence of inversions in the Vs velocity profiles (Di Giacomo *et al.* 2005). Finally, Frankel *et al.* (2002) and Park and Hashash (2004) illustrated the role of amplification from the deeper (> 30 m) sediments.

We re-analyze the empirical bases of the correlation between Vs30 and amplification and see that the relationship cannot be studied through standard regression because both variables have nonnegligible estimation errors. The use of an appropriate regression yields substantially different results. More importantly, we show that the significance of the correlation between the earthquake amplification factor F_a and Vs30 is questionable even in the original datasets used to derive it.

F_a vs. Vs30: IS THERE A RELATION?

The need for a correlation between the site-dependent earthquake amplification factor and any soil property (in this case the Vs30) for practical applications is so strongly felt that it may sometimes lead researchers to graphically stretch data to reach apparently better results. Let us re-analyze the data used by Borcherdt (1994), which is the reference work in which F_a-Vs30 relations are computed as a function of the size of the input ground motion and period classes (short, intermediate, long, and mid). Using figure 6 (Borcherdt 1994) as our reference, we resample the points with coordinates (Vs30, F_a) for the four period classes.

First of all, let us consider the linear relation between F_a and Vs30 for the short period class (0.1–0.5 s, 10–2 Hz), which is probably the most important one since it involves the fundamental vibration mode of one- to eight-story buildings, which are the most common in many countries. Borcherdt (1994) himself observed that the correlation for this class was lower than for the others.

Figure 1A shows short period data plotted on a linear scale. The scatter is so large that one could even conclude that the maximum amplification on moderately stiff soils (Vs30 about 300–350 m/s) is higher than on very soft soils. Forgetting the principle of basic data analysis (see, *e.g*, Worthing and Geffner 1943) that log-log scale should be used only if the data span several orders of magnitude, and replotting the data on a log-log scale (figure 1B), does not change the results: the scatter is too large to allow any conclusion. However, if one forgets the other principle of basic data analysis that scale should never be extended beyond the data range, and stretches the vertical scale (amplification) to the range 0.1–100 to cover three orders of magnitude instead of the less than one that is actually spanned by the data, the eye falsely perceives a linear trend that does not exist (figure 1C). This "optical effect" is enhanced if regression lines are added (figure 1D).

The same problem occurs, although less dramatically, with the other three period classes (intermediate, long, and mid).

According to elementary statistics, the quantitative estimator of the "goodness" of the linear relation between F_a and Vs30 is the standard R^2 regression coefficient. In this case (figure 1), R^2 is practically zero (0.09) for the short period class and of little significance for the other three classes (about 0.3, meaning that only 30% of the data are explained by a linear relation). However, we will show that such an elementary reasoning does not apply in this case.

F_a vs. Vs30 LINEAR RELATION: HOW SHOULD IT BE CALCULATED?

Assuming that a linear relation truly existed between F_a and Vs30, how should it be calculated? The standard linear least

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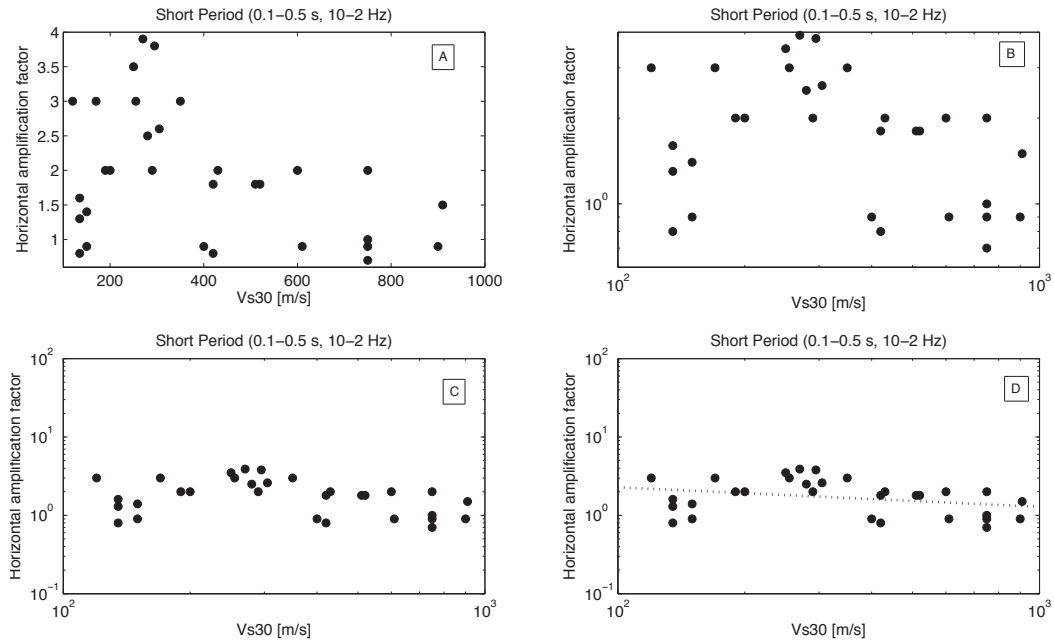


▲ **Figure 1.** Horizontal amplification factor vs. Vs30 in the short period band (0.1–0.5 s) derived from the Loma Prieta (1989) earthquake (Borcherdt 1994). (A) linear scale, (B) log-log scale, (C) stretched log-log scale, (D) stretched log-log scale with the addition of a regression line.

squares regression is not suitable because not all the requirements for its application are satisfied. One particular requirement that fails to be met is that the uncertainty on the independent variable should be at least one order of magnitude smaller than the uncertainty on the dependent variable.

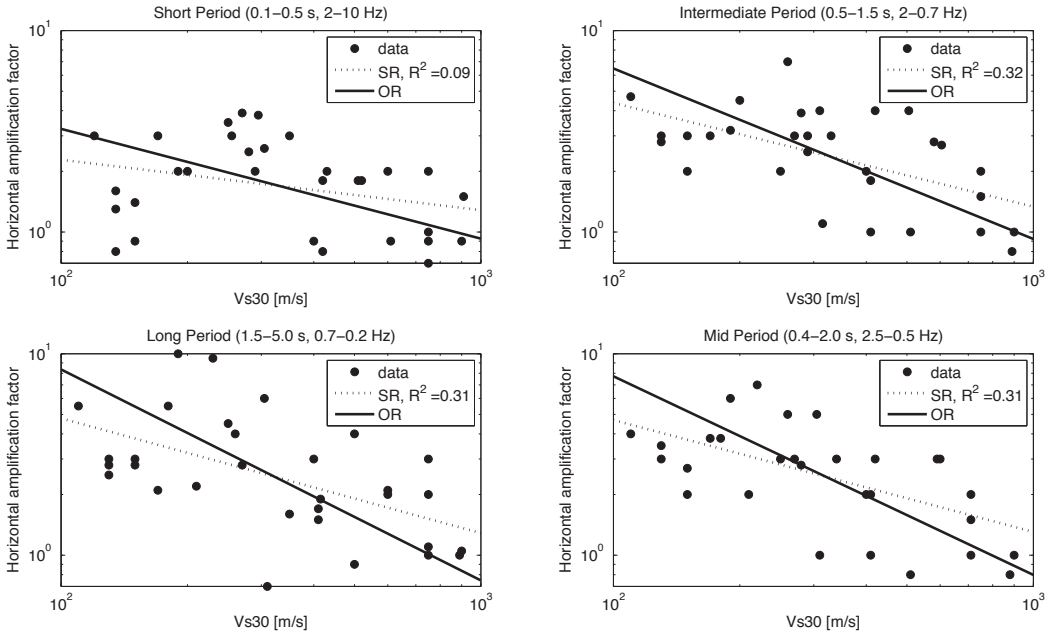
In our case both the spectral amplification and the Vs30 parameter are affected by nonnegligible experimental errors. The Vs profile can in fact be measured through direct seismic methods (cross-hole, down-hole, seismic cone) or, at lower cost, through indirect surface techniques such as active arrays or pseudoarray (SH-refraction, SASW, MASW, FTAN), passive arrays (SPAC, ESAC, ReMi), *etc*. or even derived from empirical relations with other geotechnical parameters such as the number of blows in a standard penetration test or the undrained cohesion. All these techniques have an intrinsic inaccuracy, which ranges from several percent to several tens percent.

In light of this, the regression between F_a and Vs30 cannot be computed through standard regressions but requires a regression that takes into account errors on both variables (*cf.* Castellaro *et al.* 2006; Castellaro and Bormann 2007).

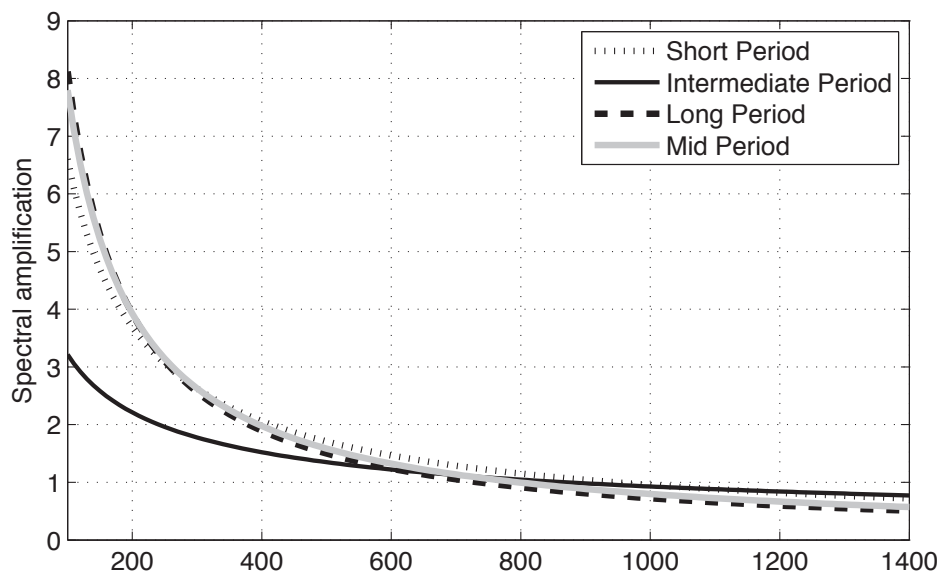
We applied orthogonal regression to the datasets assuming that errors on the dependent and the independent variables are of the same order of magnitude. In figures 2 and 3, the results of this regression are compared to those of the standard regression published in Borcherdt (1994, figures 6–7). It can be seen that the use of the standard regression would underestimate amplification by nearly a factor of two at low Vs30 values (*e.g*, for the long period class, where a Vs30 = 100 m/s would give $F_a = 4.5$ with the standard regression but would give $F_a > 8$ with the orthogonal regression). However, before considering a systematic correction of the relation between F_a and Vs30, we must establish that there is a significant relation between them.

In this case, R^2 cannot be used as a measure of the percentage of variance explained by the model because standard regression does not apply. A viable estimator of the correlation is the r' Spearman's rank correlation statistic, which respectively gives us −0.29, −0.57, −0.56, and −0.52 for the four period classes. Hence, the null hypothesis H0 : $r' = 0$ (no correlation) with the alternative of a significant correlation H1: $r' \neq 0$, tested for $(n - 2)$ degrees of freedom, yields that F_a and Vs30 are not correlated in the short period data set. For the other period classes a weak correlation is apparent at significance levels between 1% and 5%. Recalling that Spearman's correlation is conservative, this stands for a weak correlation between F_a and Vs30 even on the data on which it was originally proposed.

It should also be observed that linear regression requires far more care than common practice and that wrong results are often reached in seismology (see, *e.g*, Mulargia and Gasperini 1995). Various effects, such as nonconstant dispersion, different regimes, nonrandomness of the residuals, etc. should be carefully scrutinized and treated. In this case, for example, because Vs30 > 600 m/s often indicates rock sites (Boore and Joyner 1997) with $F_a \approx 1$, the Vs30-F_a plots will necessarily show a constant distribution for Vs30 > 600 m/s. This has two consequences: first, the trend relating Vs30 and F_a, provided that it exists, cannot be taken from Vs30 to ∞ but must at least be split into two domains with a change point (*cf.* Mulargia and Tinti 1985); similarly, even extremely highly dispersed datasets like those in Borcherdt (1994) or Harmsen (1997) may still have apparently high correlation coefficients, due to the presence of a few data with large Vs30 and low amplification. Indeed, the



▲ **Figure 2.** Horizontal amplification factor vs. Vs30 for different period bands for the Loma Prieta (1989) earthquake (Borcherdt 1994). Standard linear regression (dashed), its regression coeffcient R^2, and orthogonal regression (solid) line are added.



▲ **Figure 3.** Apparent (nonsignificant, see text) relations between Fa and Vs30 derived from the application of the orthogonal regression to the datasets of figure 2.

simple correlation analysis alone is not sufficient to explain the goodness of a regression model (*cf.* Draper and Smith 1981).

CONCLUSIONS

In spite of its almost universal adoption as a key parameter in seismic site classification, Vs30 appears a weak proxy to seismic amplification. A reassessment of the original data on which it was derived confirms this weakness.

It is unquestionable that the shear-wave subsoil velocity profile should have a role in the site effect assessment, since Vs is tied to soil rigidity. However, seismic amplification is evidently too complex to be related to a single synthetic parameter measured in the first 30 m of subsoil (Cranswick *et al.* 1990; Harmsen 1997; Hartzell *et al.* 2001; Frankel *et al.* 2002; Hartzell *et al.* 2003).

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